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Dear Hiring Committee,

I am writing to express my strong interest in the Principal Scientist position in Early Translational Informatics and Predictive Sciences – Neuroscience at Bristol Myers Squibb. As a computational neuroscientist with deep expertise in high-dimensional biomarker analysis, I bring a translational mindset that's already central to my work. My research focuses on modeling spontaneous thought—particularly repetitive negative thinking—and identifying neurobiological markers relevant to diagnosis and intervention in neuropsychiatric disorders. This role at BMS aligns naturally with my experience and aspirations, and I'm excited to contribute to your mission through rigorous, clinically meaningful science.

At Drexel, I led and co-authored studies combining neuroimaging, electrophysiology, and physiological monitoring to identify the neural and bodily signatures of spontaneous cognitive processes—such as mind-wandering, rumination, and arousal. This work uncovered candidate biomarkers, including dynamic default mode network activity and heartbeat-evoked potentials, and applied advanced mixed-effects and predictive modeling to examine how these features track with repetitive negative thinking in both healthy and at-risk populations. These experiences have deepened my appreciation for the challenges of identifying reproducible, clinically meaningful brain-behavior associations in heterogeneous samples—a perspective I'm eager to bring to translational biomarker efforts at BMS.

I am highly proficient in R and Python, with expertise in multivariate modeling, dimensionality reduction, and the integration of behavioral, physiological, and self-report datasets. My approach to exploratory data analysis is both rigorous and collaborative, and I'm particularly motivated by the challenge of understanding individual differences in treatment response and disease progression. What excites me most about this role is the opportunity to apply these skills toward identifying therapeutic targets, informing clinical trial enrichment strategies, and advancing precision biomarkers for neuropsychiatric and neurodegenerative conditions. I believe my translational mindset, technical fluency, and neuroscience background would make me a strong asset to your team.

Thank you for considering my application. I would welcome the opportunity to contribute to BMS's pioneering work at the intersection of computational science, human biology, and translational neuroscience, and I look forward to the chance to speak with you further.

Sincerely,

David Braun, Ph.D.